A FINAL PROJECT REPORT ON

LANDSLIDE SUSCEPTIBLITY MODELLING USING MACHINE LEARNING METHODS

SUBMITTED BY

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CERTIFICATE

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ABSTRACT

Landslide is a natural hazard that results in many economic damages and human losses every year. Numerous researchers have studied landslide susceptibility mapping (LSM), each attempting to improve the accuracy of the final outputs. In this project I have analysed a case study from **Longzu**, Three Gorges reservoir area in **china** and Landslide Susceptibility assessment of Uttarakhand region of India.
Machine Learning methods that have been employed for susceptibility assessment and prediction are
Support vector machines (SVM)
Artificial Neural Network(ANN)
multivariate statistical model ,Logistic Regression(LR)

Different methods were implemented to define the landslide locations, such as inventory reports, satellite images and field survey. Moreover, **14** landslide conditioning factors were considered in the analysis of landslide susceptibility. Later, the unimportant factors were selected and eliminated using the **information gain ratio** technique. The landslide locations were randomly divided into two groups: 70% for training and 30% for verifying Results show that the elimination of noise-generating factors and the separated modelling of each landslide type have significantly increased the **prediction accuracy**. The machine learning models outperformed the multivariate statistical model and **SVM** model was found ideal for the above case study area.

1.Brief Description about landslide and factors affecting its occurrence.

1.1Defintion

The terms landslide or mudslide refer to the downward movement of large masses of rocks, soil, mud and organic debris. Areas with steep slopes, for example mountainous regions, are particularly susceptible to landslide hazards.

Landslides are mainly caused by gravity acting on weakened rocks and soil that make up a sloping area of land. Both natural and human-related activities can increase the risk for landslides. Water from heavy rainfall is a frequent trigger for landslides because it’s heavy and adds extra weight to the land.

1.2 Causes

Landslides occur when the slope (or a portion of it) undergoes some processes that change its condition from stable to unstable. This is essentially due to a decrease in the shear strength of the slope material, to an increase in the shear stress borne by the material.

Natural causes of landslides include:

- saturation by rain water infiltration, snow melting, or glaciers melting.
- increase of hydrostatic pressure in cracks and fractures
- erosion of the toe of a slope by rivers or ocean waves;
- physical and chemical weathering (e.g. by repeated freezing and thawing, heating and cooling)
- ground shaking caused by earthquakes, which can destabilize the slope directly
- volcanic eruptions;

Landslides are aggravated by human activities, such as:

deforestation, cultivation and construction,blasting and mining, vibrations from machinery or traffic.

1.3 Factors taking part in Landslides occurence

Fourteen conditioning factors were considered in the analysis of landslide susceptibility based on the literature review and data availability.

Topographic factors that take part in LSM datasets are

Slope

Slope has considerable influence on slope stability, which can be explained as the form between any section of the surface and a horizontal datum. Many studies have shown that slope is a main conditioning factor in LSM

Aspect

Aspect related factors including exposure to sunlight, winds and precipitation are vital factors in instigating landslides

Altitude

Altitude is affected by several geomorphologic and geological processes, and thus, it is an influential conditioning factor in landslide analyses

Curvature (plan and profile)

Plan curvature represents the flow acceleration and erosion/deposition rate. Profile curvature affects the flow velocity variation down the slope

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Sediment transport index (STI)

STI represents the slope failure and deposition procedures (Moore and Wilson 1992 Moore ID, Wilson JP. 1992. Length-slope factors for the revised universal soil loss equation: simplified method of estimation. There is a direct relationship between the STI and the water accumulation at the bottom of the catchment as well as the erosion amount

$$STI = \left(\frac{AS}{22.1}\right) 0.6 \sin\left(\frac{\beta}{0.089}\right) 1.3$$

where β denotes the slope at a certain pixel and AS refers to the upstream area.

Topographic wetness index (TWI)

TWI is described by the flow accumulation theoretical measure, where soil moisture affects the slope material, which results in pore water pressure and diminishes soil stability , for landslide susceptibility mapping by and could be regarded as an influential factor in landslide studies.

$$\text{TWI} = \log\left(\frac{A}{b tan\beta}\right)$$

where A denotes flow accumulation in square meters, b refers to the pixel width through which water flows in meters and β (radian) represents the slope

Terrain roughness index (TRI)

TRI shows the concave and convex upward slopes

$$\text{TRI} = |x|\sqrt{(max^2 - min^2)}$$

Where max denotes the pixel largest value in nine rectangular altitude neighbourhoods, and min denotes the minimum value.

Stream power index (SPI)

SPI represents flowing water erosion power according to the presumption which relates discharge to definite catchment area.

SPI = A*tanβ/b where A is the specific catchment area and β is the local slope gradient computed in degrees.

Buffer zones (distance to fault and stream)

The landslides occurred mainly along the fault and decreased sharply with distance from it. A buffer map related to the seismogenic fault with a 5-km buffer zone interval .Moreover, the distance to the stream was derived in two steps. First, the streams were delineated using flow accumulation and converted into a vector format. In the second step, a Euclidean distance analysis was conducted to produce the distance to the stream factor.

Land use

Land use is affected by human activities and alterations in the environment These classes include agriculture, dry farming, high density forest, medium density forest, mixed forest/orchard, mixed agriculture/orchard, orchard, mixed orchard/agriculture, sandy/dune and urban.

Lithology

Lithology can be regarded as one of the most vital conditioning factors in LSM because strength and the rocks and soils permeability are influenced directly by lithological characteristics

Bedding structure

Bedding structure indicates the intersection relationship between strata and slope.

Distance to faults

The proximity parameters (distance to faults, rivers and roads) were calculated using geological and geomorphology maps based on the Euclidean distance .most of the landslides occurred far away from the faults. Within the influence area, the faults show a more positive effect on landslide occurrence.

2. Methodology

2.1 Information value model

The information value model (Yin and Yan, 1988) is based on the concept that landslide occurrence (y) is affected by various factors (xi), and their influences to landslides are different. According to a conditional probability, the formula for the information value can be written as:

$$I(y,x) = \frac{P(y|x)}{P(y)}$$

Where $I(y,x)$ is the information value under the causal factors x ;$P(y)$ is the probability of landslide occurrence; $P(y|x)$is the probability of the occurrence of landslide under the causal factor x.

2.2 Information gain ratio

Information gain ratio (IGR) is one of the most efficient feature selection methods . The factors with a higher value of IGR indicate a higher prediction ability of the models. Assuming that the training data T consists of n samples, and belongs to the class Ci (landslide, non landslide). Then, the information entropy can be calculated as:

$info(T) = -\sum_{i=1}^{2}(n(Ci,T)\backslash\ |T|) \ \ * \log(\frac{n(Ci,T)}{|T|})$

The amount of information(T_1,T_2,,,….T_m) split from T regarding the casual factor F is estimated as:

$$info(T,F) = -\sum_{j=1}^{m} \frac{T_J}{|T|} \log\big(info(T)\big)$$

3. Landslide susceptibility modelling

3.1 Support Vector Machines

Support vector machine (Vapnik, 1995) is a The input variables in the original space are mapped into a high-dimensional linear feature space by nonlinear transformation. Then, in order to split the positive from the negative, SVM model operates by attempting to find an optimal surface in the feature space between the two types nonlinear classification method.

Let x, y for i =1,2,3,….n be n samples,the optimal hyper plane eqn can be obtained by:

$$Min\left(\frac{1}{2}\ ||\boldsymbol{w}||^2 + C\sum_{i=1}^{n}\xi\right)$$

$$y(w.x + b) - 1 + \xi \geq\ 0,\ \ \xi \geq 0, i = 1,2, \dots n$$

Where **w**is the weight vector that determines the orientation of the hyperplane, b is the bias, zeta(ξ)is the positive slack variables for the data points that allow for penalized constraint violation, C is the penalty parameter that controls the trade-off between the complexity of the decision function and the number of training examples misclassified.

3.2 Artificial neural Network

Artificial neural network is a reasoning model established on the imitation of human brain function and nervous system. Back propagation neural network (BPNN) is one of the most effective ANNs, it is a multilayer neural network consisting of an input layer, hidden layers, and

an output layer . In signal propagation, the input signal is processed layer by layer from the input to the output. If the result of the output layer is not expected, it would be transferred to the reverse propagation, and adjust to the network weights and thresholds according to the prediction error to approximate the desired output.

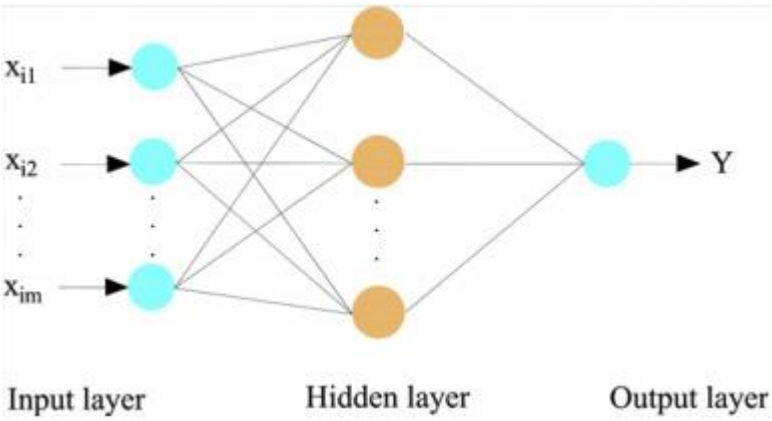


FIG: (The architecture of a three layers BPNN)

The learning rate is an important parameter of ANN model, which may affect its performance. In this study, the learning rate will be automatically calculated using the following formula:

$$\eta(n) = \eta(n-1) * \exp(\log \eta_max/\eta_\min)/d$$

Where η is the learning rate in the nth times training; η_min is the minimum value of the learning rate; η_max is the maximum value of the learning rate, and d is the delay rate. In this study, the initial rate, the maximum and minimum learning rate, and the delay rate are 0.3, 0.1, 0.01 and 30, respectively.

3.3 Logistic Regression

Logistic regression is a multivariate statistical method for landslide susceptibility mapping . LR can reveal the relationship between a target variable and multiple predictor variables, and predict the occurring probability of a certain event. In a statistical analysis of LR, the predictor variables can be either continuous or discrete, and there is no need to meet the normal distribution. LR is well suited in describing and validating the relationship between categorical outcome variables (landslide or non-landslide) and categorical or continuous

predictor variables (landslide affecting factors). In the case of landslide prediction, the main purpose of LR is to find the best fitting algorithm to analyze the spatial relationship between the presence or absence of a landslide event.

The logit-natural logarithm of LR is indicated by an equation of the form:

$$Y = f(z) = \frac{1}{1 + \exp(-Y)}$$

Y = $\boldsymbol{X\theta}$ **,** X is a feature vector and theta(θ) is corresponding weight associated.

In terms of probability(P)The prob. of landslide occurnce is P,

$$Y = f(P) = \ln(\frac{P}{1-P})$$

$$P = p(Y|X) = \frac{\exp(X\theta)}{1 + \exp(X\theta)}$$

where Y represents outcome variables (landslide or non-landslide), X = X1,X2,…,Xn represents predictor variables of landslide affecting factors, n is the number of landslide affecting factors.

4. STUDY AREA

A comparative study of different machine learning methods for landslide susceptibility assessment:

A case study of Uttarakhand area (India)

Description of the study area

The study area is located between longitudes 78°37′40′E to 79°00′50″E and latitudes 30°23′15″N to 30°03′58″N in part of the Uttarakhand state of India, covering about 1325 km2 (Fig 1). This area is situated in a subtropical monsoon region having annual average rainfall of 600 mm. Heavy rainfalls usually take place in monsoon season (June to September). Temperature during the year ranges from below 0 °C during winter to 45 °C during summer. Mean relative humidity var ies from 25% to 85%.

Topographically, the study area belongs to mountainous region with intervening valleys**.** Elevation in the area varies from 450 m to 2738 m above sea level. Slope in this area is steep having slope angles up to 70°. Geologically, the area is structurally complex having folded igneous, sedimentary and metamorphic rocks (Agarwal and Kumar, 1973). Metamorphic rocks of Jaunsar group (phyllite and quartzite) are occupying major part of the area. Predominant soil types in the area are silty and loamy

Preparation of geospatial database.

For landslide prediction, the spatial relationship between historical landslide events and geo-environmental factors must be carried out.Based on the mechanism of landslide occurrences and characteristics of geo-environment of the study area, a set of fifteen landslide conditioning factors

(slope angle, slope aspect, elevation curvature, plan curvature, profile curvature, soil type, land cover, annual rainfall, distance to lineaments, distance to roads, distance to rivers, lineament density, road density, and river density)

have been taken into account for initial landslide analysis.

Landslide affecting factors have been classified into different classes .This classification is based on the analysis of susceptibility of each factor class to landslide occurrences carried

No.	Landslide affecting factors	Class
1	Slope angle (degree)	(1) [0,10); (2) [10,20); (3)[20,30); (4) [30,40); (5) [40,50); (6) [50,60); (7) [60,69]
2	Slope aspect	1) flat; (2) north; (3) northeast; (4) east; (5) southeast; (6) south; (7) southwest; (8) west; (9) northwest
3	Elevation (m)	(1) [450,700); (2) [700,900); (3) [900,1100); (4) [1100,1300); (5) [1300,1500); (6) [1500,1700); (7) [1700,1900); (8) [1900,2100); (9) [2100,2300); (10) [2300, 2738]

No.	Landslide affecting factors	Class
4	Curvature	(1) high concave [−18.25,−2); (2) concave [−2,−0.05); (3) flat [−0.05,0.05); (4) convex [0.05–2); (5) high convex [2,24.25]
5	Plan curvature	(1) [−8.559,−1.078); (2) [−1.078,−0.344); (3) [−0.3444,0.257); (4) [0.257,0.992); (5) [0.992–8.472]
6	Profile curvature	(1) [−16.25,−1.385); (2) [−1.385,−0.433); (3) [−0.433,0.280); (4) [0.280,1.232); (5) [1.232–13.956]
7	Soil type	(1) coarse-loamy; (2) fine-loamy; (3) skeletal-loamy; (4) mixed-loamy; (5) fine-silt
8	Land cover	(1) dense forest; (2) open forest; (3) scrub land; (4) arable land
9	Annual rainfall (mm)	(1) [200,300); (2) [300,400); (3) [400,500); (4) [500,600); (5) [600,700); (6) [700,800); (7) [800,1002]
10	Distance to lineaments (m)	(1) [0,100); (2) [100,200); (3) [200,300); (4) [300,400); (5) [400,500); (6) [500,600); (7) [600,700); (8) ≥ 700
11	Distance to roads (m)	(1) [0,50); (2) [50,100); (3) [100,150); (4) [150,200); (5) [200,250); (6) ≥ 250
12	Distance to rivers (m)	(1) [0,50); (2) [50,100); (3) [100,150); (4) [150,200); (5) [200,250); (6) ≥ 250
13	Lineament density (km/km$^{2)}$	(1) very low [0,0.673); (2) low [0.673,0.939); (3) moderate [0.939,1.184); (4) high [1.184,1.469); (5) very high [1.469,2.602]
14	Road density (km/km$^{2)}$	(1) very low [0,0.05); (2) Low [0.05,0.154); (3) moderate [0.154,0.268); (4) high [0.268,0.423); (5) very high [0.423,1.268]
15	River density (km/km$^{2)}$	(1) very low [0,0.086); (2) low [0.086,0.589); (3) moderate [0.589,1.049); (4) high [1.049,1.623); (5) very high [1.623,3.663]

Model construction using machine learning algorithms

Support Vector Machine(SVM)

It is based on the statistical approach in order to find an optimal hyper-plane for separating two classes Let X = (X1, X2, …, Xn) is the vector of fifteen landslide affecting factors, and Yj = (Y1, Y2) is the vector of classified variables (landslide and non-landslide). The optimal separating hyper-plane can be obtained by solving the classification function as below:

$$f(X)=\mathrm{sign}[\sum_{i=1}^{n}\alpha_i Y_j k(X,X_i)+c]$$

where c is the offset from the origin of the hyper-plane, n is the number of landslide affecting factors, αi are positive real

constants,k(X,Xi) is the kernel function that can be linear, polynomial, radial basis function classification (landslide, non-landslide), the condition for solving Equation (1) is assumed as below:

$$(Y_j[\omega^T\varphi(X_i)+c]\geq 1\Leftrightarrow$$

$$\{\omega^T\varphi(X_i)+c\geq 1, \text{if} Y_j=+1(\text{landslide})$$

$$\omega^T\varphi(X_i)+c\leq 1, \text{if} Y_j=-1(\text{non-landslide}$$

Logistic regression (LR)

It is based on the central mathematical concept of the logit-natural logarithm. LR is well suited in describing and validating the relationship between categorical outcome variables (landslide or non-landslide) and categorical or continuous predictor variables (landslide affecting factors). In the case of landslide prediction, the main purpose of LR is to find the best fitting algorithm to analyze the spatial relationship between the presence or absence of a landslide event and a set of affecting factors . The logit-natural logarithm of LR is indicated by an equation of the form:

$$Y=f(P)=\ln(P_1-P)=\beta_0+\beta_1X_1+\beta_2X_2+\ldots+\beta_nX_n$$

Therefore, the probability of a landslide event (P) can be determined from following equation:

$$P=P(Y|X)=\frac{e^{\beta_0+\beta_1X_1+\beta_2X_2+\ldots+\beta_nX_n}}{1+e^{\beta_0+\beta_1X_1+\beta_2X_2+\ldots+\beta_nX_n}}$$

where Y represents outcome variables (landslide or non-landslide), X = X1,X2,…,Xn represents predictor variables of landslide affecting factors, n is the number of landslide affecting factors, β0 is theintercept condition, β1,β2,…,βn are the regression coefficients.

Bayesian Network (BN)

It is based on Bayes' theorem to represent graphically and probabilistically correlative and causal relationships among variables BN is usually used widely in modeling of complex systems (Song et al., 2012b). For landslide prediction, BN is utilized to analyze the affecting factors contributing to rainfall-induced landslides from which assessing the susceptibility of landslide occurrences. The unique joint probability of a landslide event in relation with a set of affecting factors using BN can be estimated by equation of below form:

$$P_B(X_1,X_2,\ldots,X_n)=\prod_{i=1}^{n}P_B(X_1||\prod X_i)=\prod_{i=1}^{n}\theta_{X_i|\prod X_i}$$

where X = (X1,X2,…,Xn) represents landslide affecting factors, PB(X|∏Xi)=θXi|∏Xi is a joint probability distribution over affecting factor Xi, n is the number of landslide affecting factors.

Naïve Bayes (NB)

NB is also based on Bayes' theorem (Pham et al., 2015b) that is a statistical approach using a conditional independence assumption that all attributes are independent with given output class . The main purpose of NB is to determine the prior probability of an event based on the proportion of the observed cases in relation with given output class. For spatial prediction of landslides, suppose that X = (X1,X2,…,Xn) is the vector of landslide affecting factors, Yj = (Y1, Y2) is the vector of categorical variables (landslide or non-landslide). The prior

probability of a landslide event in relation with a set of affecting factors using NB can be estimated from following equation:

$$Y_{NB}=\underset{Y_i=[landslide,nonlandslide]}{argmax}P(Y_i)\prod_{i=1}^{n}P(X_i/Y_i)$$

where, P(Yi) is the prior probability of event Yi, P(Xi/Yi) is the conditional probability calculated as below:

$$P(X_i/Y_i)=\frac{1}{2\pi\sqrt{\delta}}e^{-\frac{(X_i-\eta)^2}{2\delta^2}}$$

where δ is the standard deviation of Xi, η is the mean value of Xi.

According to Tien Bui et al. (2015), the performance of landslide models should be evaluated and compared for both training and testing datasets. For training dataset, the results show the degree of fit of landslide models over the data or the degree of fit between input and output using landslide models. Meanwhile, using testing dataset, the results show the predictive capability of landslide models (Tien Bui et al., 2012a). In this study, statistical index-based evaluations and ROC curve have been utilized to evaluate and compare the performance of five landslide models.

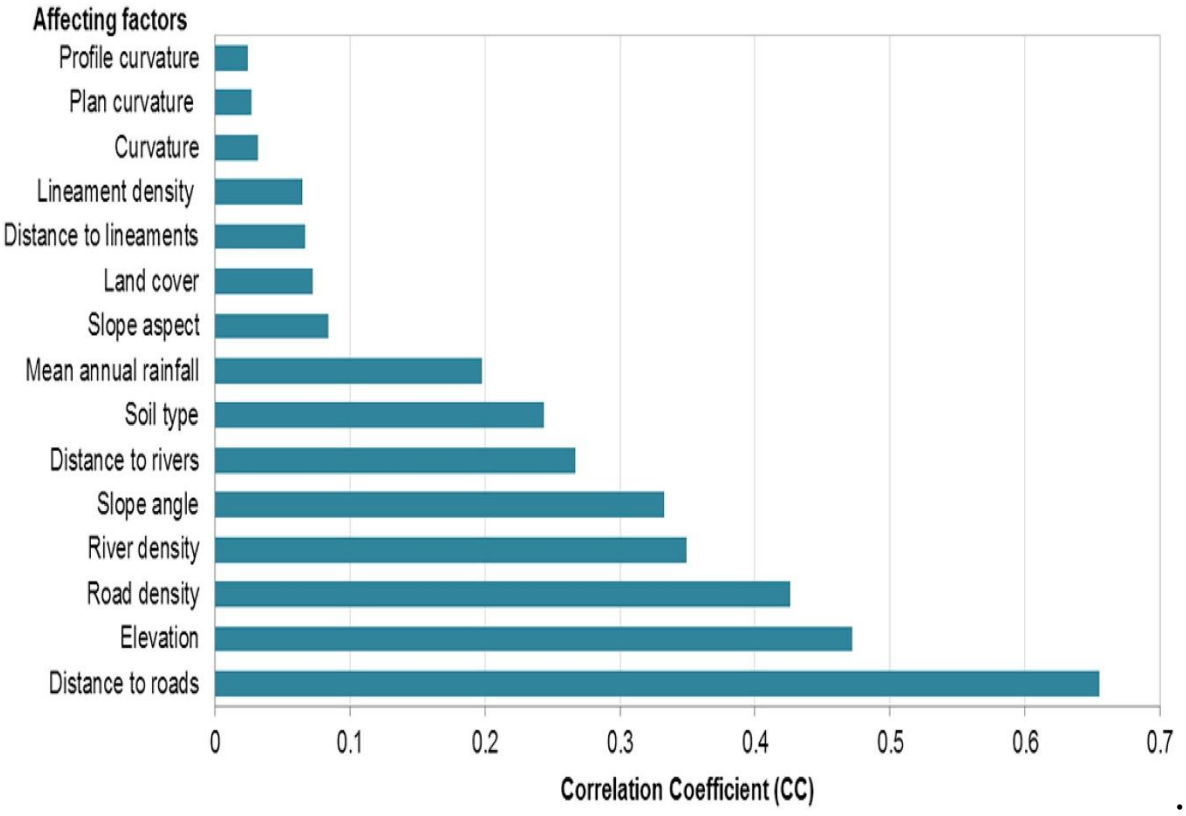
Statistical index based evaluations

Model study and analysis

Geo-environmental factors (slope angle, slope aspect, elevation, curvature, plan curvature, profile curvature, soil type, land cover, mean annual rainfall, distance to lineaments, distance to roads, distance to rivers, lineament density, road density, and river density) have been considered as landslide affecting parameters in the present study. However, the contribution of these factors to landslide models might be different. Thus, it is necessary to evaluate the predictive capability of these landslide affecting factors to select suitable factors and remove irrelevant or unimportant factors for further analysis.

Correlation based Feature Selection (CFS), which is known as one of the most efficient feature selection methods (Hall, 1999), has been utilized in this study to evaluate the predictive capability of landslide affecting factors. CFS is carried out based on the correlation between each factor and each class, and inter-correlation between factors (Nguyen et al., 2010). The factors with higher Correlation Coefficient (CC) values indicate higher contribution to landslide models and vice versa.. Based on these results, it can be concluded that all

fifteen landslide affecting factors are having contribution to landslide models (CC > 0). However, there is no arguments to remove factors with low CC values such as curvature, plan curvature, profile curvature. Therefore, Backward Elimination (BE) method has been utilized further in the present study to provide more evidence in removing irrelevant or unimportant factors to improve the performance of landslide models as the elimination method helps in finding the most effective features for landslide modelling

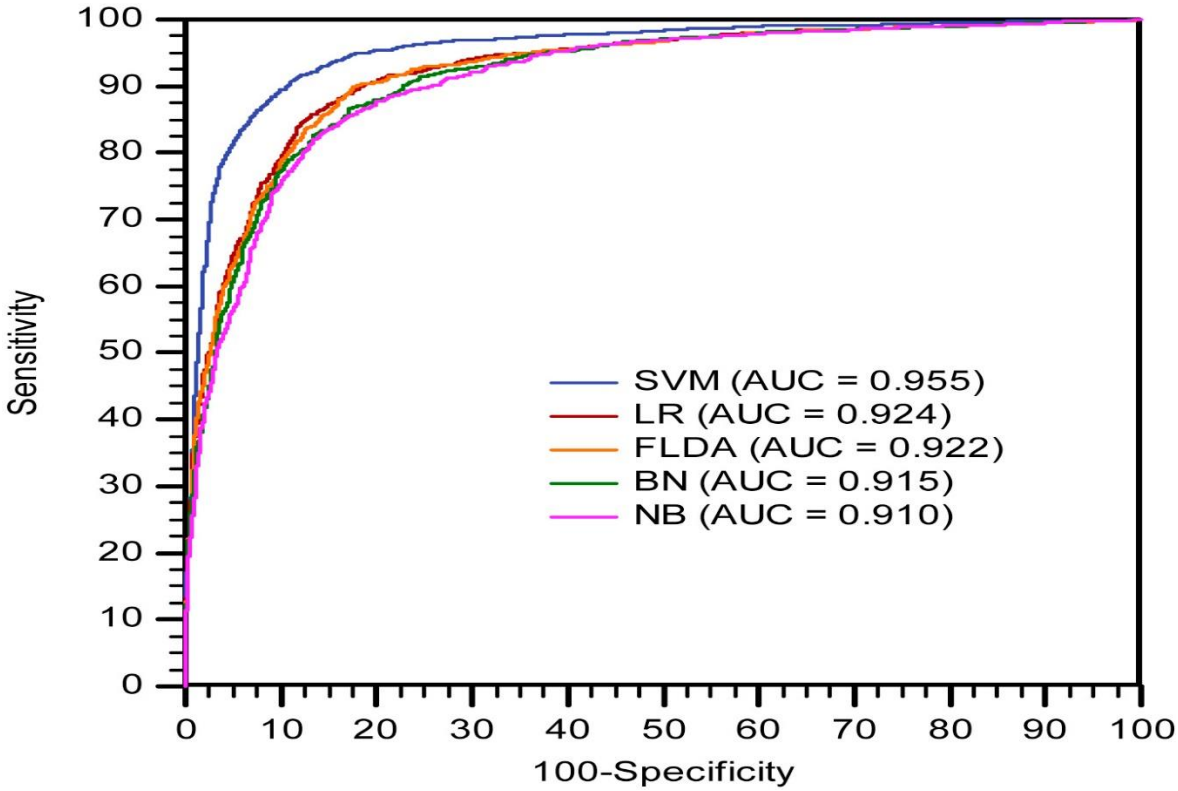


Based on above analysis of selection of affecting factors, out of fifteen, only nine landslide affecting factors have been selected for landslide modeling in the present study. These factors are slope angle, slope aspect, elevation, soil type, distance to roads, distance to rivers, lineament density, road density, and river density.

Training and validating the landslide model

non-landslide pixels have also been selected randomly from non-landslide area. For training dataset, 2924 non-landslide pixels have been used, and 1253 non-

landslide pixels have been utilized to construct testing dataset. Landslide non-landslide pixels with grid size of 20 × 20 m have been sampled with landslide affecting factors to generate the final datasets. Using training dataset, landslide models (SVM, LR, FLDA, BN, and NB) have been constructed.The landslide inventory has been first divided randomly into two parts having. First part has been used to generate training dataset, and second part has been utilized to construct testing dataset. Similarly,



Model Performance analysis

Landslide susceptibility maps developed from different landslide models have been validated by calculating Landslide Density (LD) of each susceptible class on the maps.LD is a ratio of the percentage of landslide pixels (P_{LS}) and the percentage of all pixels (P_P) of each susceptible class on the map as shown in following equation

$$\mathbf{LD} = \frac{\boldsymbol{Pls}}{\boldsymbol{Pp}}$$

Class	LD			
	SVM	**LR**	**BN**	**NB**
Very low	0.02	0.05	0.04	0.05
Low	0.04	0.06	0.06	0.08
Moderate	0.06	0.13	0.08	0.31
High	0.13	0.42	0.28	0.94
Very high	4.69	4.76	3.21	7.66

Discussions and conclusions

In the present study, five machine learning methods namely SVM, LR, FLDA, BN, and NB have been evaluated and compared for landslide susceptibility assessment. Two of these methods namely SVM and LR has already been applied widely in landslide predictionAnalysis of comparative results indicated that out of these five models, the SVM model is one of the best methods. This has also been observed by other landslide studies in the present study, the LR model using a sequence of convergence criterions in maximizing the likelihood function for classification that can improve its performance has shown good predictive capabilityThe results also show that the BN model outperforms the NB model because BN can also indicate uncertainty among random p, and it provides a semantic way for handling missing data and avoiding over-fitting problems.

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